The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 November 1999

INCH-POUND

MIL-PRF-19500/525B
10 August 1999
SUPERSEDING
MIL-S-19500/525A
30 March 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6546, 2N6547, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (T0-3).

1.3 Maximum ratings.

Limits	P_T 1/		V_{CEX}	V_{CEO}	V_{EBO}	I_B	I_C	T_{OP} and T_{stg}	$R_{\theta JC}$
	T_C = +25°C	T_C = +100°C							
	W	W	V dc	V dc	V dc	A dc	A dc	°C	W/°C
2N6546	175	100	600	300	8	10	15	-65 to +200	1.0
2N6547	175	100	850	400	8	10	15	-65 to +200	1.0

1/ Between T_C = +25°C and T_C = +200°C, linear derating factor (average) = 1.0 W/°C.

1.4 Primary electrical characteristics.

Limits	h_{FE2} 1/	h_{FE3} 1/	C_{obo}	$\lvert h_{fe} \rvert$	Switching times	
	V_{CE} = 2 V dc I_C = 5 A dc	V_{CE} = 2 V dc I_C = 10 A dc	V_{CB} = 10 V dc I_E = 0 0.1 MHz ≤ f ≤ 1 MHz	V_{CE} = 10 V dc I_C = 1 A dc f = 1 MHz	V_{CC} = 250 V dc I_C = 10 A dc	
					t_{on}	t_{off}
			pF			
Min	12	6		6		
Max	60		500	30	1.0	4.7

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein. The preferred measurements used herein are the metric units. However, this transistor was designed using inch-pound units of measurement. In case of conflicts between the metric and inch-pound units, the inch-pound units shall rule.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition requirements (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical requirements shall be the subgroups specified in table 1 herein.

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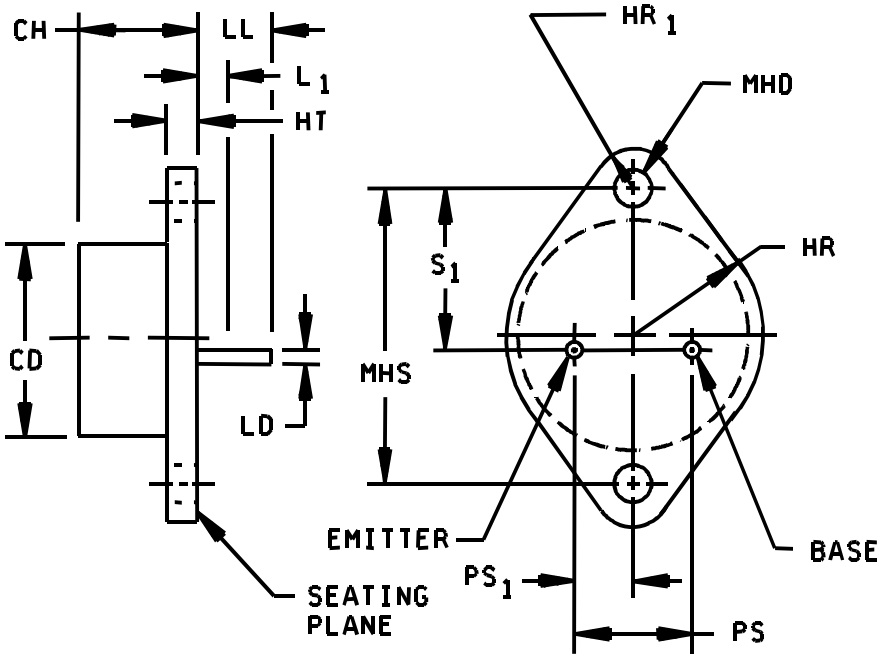


FIGURE 1. Dimensions and configuration (T0-3).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		0.875		22.23	
CH	0.250	0.450	6.35	11.43	
HR	0.495	0.525	12.57	13.34	
HR_1	0.131	0.188	3.33	4.78	
HT	0.050	0.135	1.27	3.43	
LD	0.038	0.043	0.97	1.09	
LL	0.312		7.92		
LL_1		0.050		1.27	
MHD	0.151	0.161	3.84	4.09	
MHS	1.177	1.197	29.90	30.40	
PS	0.420	0.440	10.67	11.18	3
PS_1	0.205	0.225	5.21	5.72	3
s_1	0.655	0.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points 0.050 inch (1.27 mm) and 0.055 inch (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within 0.001 inch (0.03 mm) concave to 0.004 inch (0.10 mm) convex inside a 0.930 inch (23.62 mm) diameter circle on the center of the header and flat within 0.001 inch (0.03 mm) concave to 0.006 inch (0.15 mm) convex overall.
5. Mounting holes shall be deburred on the seating plane side.
6. Collector is electrically connected to the case.

FIGURE 1. Dimensions and configuration (T0-3) continued.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
10	T_A = 150°C; t = 48 hours min. V_{CB} = 200 V dc for 2N6546 V_{CB} = 300 V dc for 2N6547
11	h_{FE2} ; I_{CEX1}
12	See 4.3.1
13	Subgroup 2 of table I herein. ΔI_{CEX} = 100 percent of initial value of 100 uA dc, whichever is greater; Δh_{FE2} = ±25 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5°C ± 12.5°C; V_{CB} ≥ 20 V dc.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, herein.

4.4.2.1 Group B inspection, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500

Subgroup	Method	Conditions
B3	1037	For solder die attach: V_{CE} ≥ 20 V dc, for 2,000 cycles, T_A ≤ 35°C.
B3	1026	For eutectic die attach: V_{CE} ≥ 20 V dc; adjust P_T to achieve T_J = 175°C minimum, T_A ≤ 35°C.
B5	3131	V_{CE} = 10 V dc, I_C = 4 A, t_H = 1 s, I_M = 20 mA dc; $R_{\theta JC}$ = 1.0 °C/W.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, herein.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 10 pounds; time = 15 seconds.
C6	1037	For solder die attach: $V_{CE} \geq 20$ V dc, for 6,000 cycles, $T_A \leq 35°C$.
C6	1026	For eutectic die attach: $V_{CE} \geq 20$ V dc; adjust P_T to achieve $T_J = 175°C$ minimum, $T_A \leq 35°C$.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination						
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6546				300		V dc
2N6547				400		V dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 8 V dc	I_{EBO}		1.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc;	I_{CEX1}		1.0	mA dc
2N6546		V_{CE} = 600 V dc				
2N6547		V_{CE} = 850 V dc				
Base emitter voltage (saturated)	3066	Test condition A; I_C = 10 A dc; I_B = 2.0 A dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		1.6	V dc
Collector to emitter saturated voltage	3071	I_C = 10 A dc; I_B = 2.0 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.5	V dc
Collector to emitter saturated voltage	3071	I_C = 15 A dc; I_B = 3.0 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		5.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE1}	15		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE2}	12	60	
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE3}	6		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A;	I_{CEX2}		30	mA dc
2N6546		V_{BE} = 1.5 V dc				
2N6547		V_{CE} = 600 V dc				
		V_{CE} = 850 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low temperature operation:		T_A = -55°C				
Forward-current transfer current	3076	V_{CE} = 2.0 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE4}	4		
Subgroup 4						
Pulse response	3251	Test condition A except test circuit and pulse requirements in accordance with figure 2 herein.				
Turn-on time		V_{CC} = 250 V dc; I_C = 10 A dc; I_{B1} = I_{B2} = 2 A dc	t_{on}		1.0	µs
Turn-off time		V_{CC} = 250 V dc; I_C = 10 A dc; I_{B1} = I_{B2} = 2 A dc	t_{off}		4.7	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = .5 A dc; f = 1 MHz	$\lvert h_{fe} \rvert$	6.0	30	
Output capacitance (open circuit)	3236	V_{CB} = 10 V dc; I_E = 0; 0.1 MHz ≤ f ≤ 1.0 MHz	C_{obo}		500	pF
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C; t = 1 s; 1 cycle; (see figure 3)				
Test 1 (Both device types)		V_{CE} = 11.7 V dc; I_C = 15 A dc				
Test 2 (Both device types)		V_{CE} = 20 V dc; I_C = 8.75 A dc				
Test 3						
2N6546		V_{CE} = 250 V dc; I_C = 45 mA dc				
2N6547		V_{CE} = 350 V dc; I_C = 30 mA dc				

See footnote at end of table.

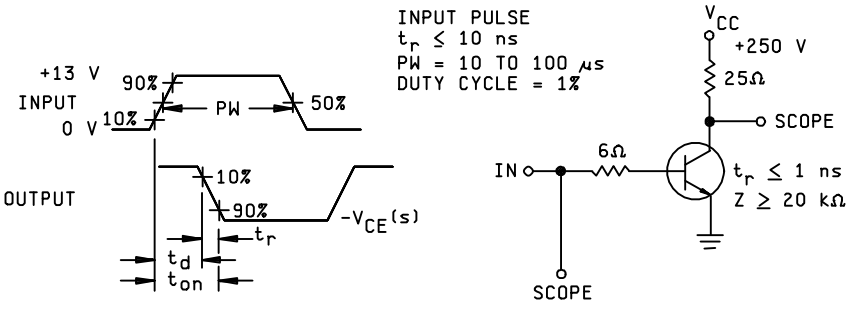
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TABLE I. Group A inspection - Continued.

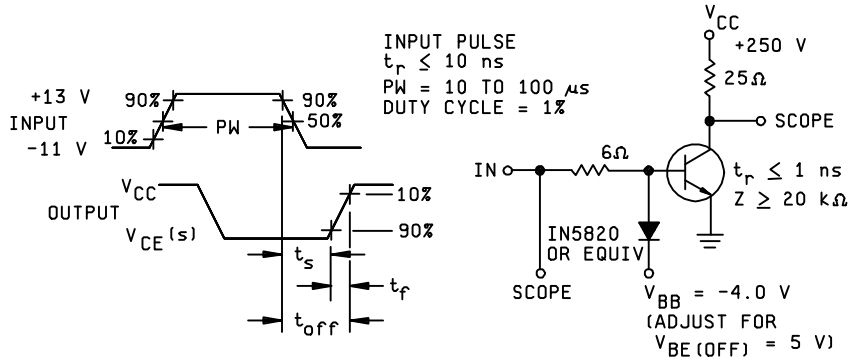
Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area	3053	Load condition C; (unclamped inductive load); See figure 4; T_C = 25°C; duty cycle ≤ 10 percent R_S = 0.1 Ω; $t_r = t_f$ ≤ 500 ns				
Test 1		t_p = 5 ms; (vary to obtain I_C); R_{BB1} = 15 Ω; V_{BB1} = 38.5 V dc; R_{BB2} = 50 Ω; V_{BB2} = -4 V dc; V_{CC} = 20 V dc; I_C = 15 A dc; L = 10 µH 2/				
Test 2		t_p = 5 ms; (vary to obtain I_C); R_{BB1} = 15 Ω; V_{BB1} = 38.5 V dc; R_{BB2} = 50 Ω; V_{BB2} = -4 V dc; V_{CC} = 20 V dc; I_C = 100 mA dc; L = 1 mH 3/				
Electrical measurements: Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc	I_{CEX3}			
2N6546		V_{CE} = 600 V dc			2.0	mA dc
2N6547		V_{CE} = 850 V dc			2.0	mA dc
Safe operating area (switching)		Clamped inductive load; T_A = 25°C; duty cycle ≤ 5 percent; t_p = 1.5 ms; (vary to obtain I_C); V_{CC} = 20 V dc; I_C = 8 A dc; L = 180 µH (see figure 5)				
2N6546		Clamp voltage = 350 V dc				
2N6547		Clamp voltage = 450 V dc				
Electrical measurements		Same as safe operating area (unclamped inductive) above				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ L = 10 µH (approx. 10 turns, 1 row of #16ASG wire on an air core 2 7/8" ID) .0007 ohms, or equivalent.
3/ L = 1 mH (1 each Miller type 7827 in parallel with 2 each series strung Miller type 7825 and this in series with 2 each series strung Miller type 7827) .45 ohms, or equivalent.

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TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 2. Switching time test circuits.

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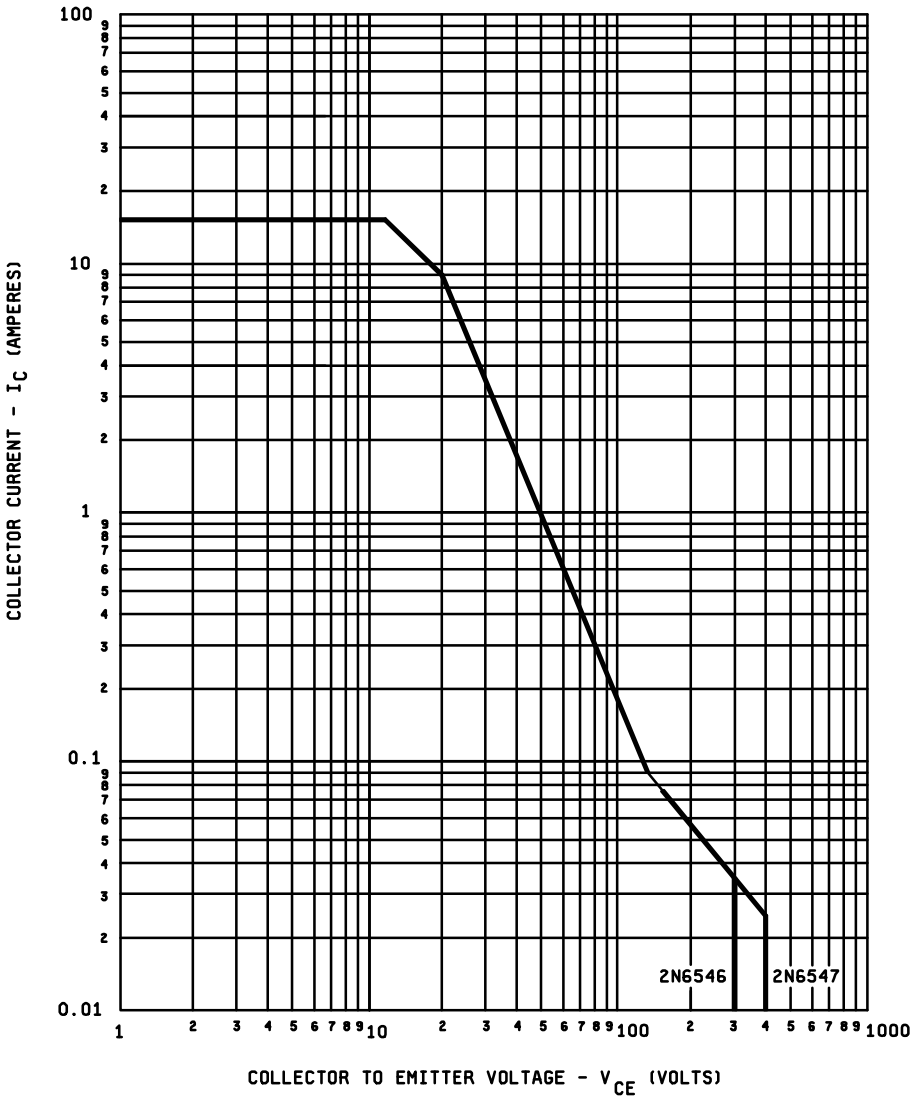


FIGURE 3. Maximum safe operating area graph (continuous DC).

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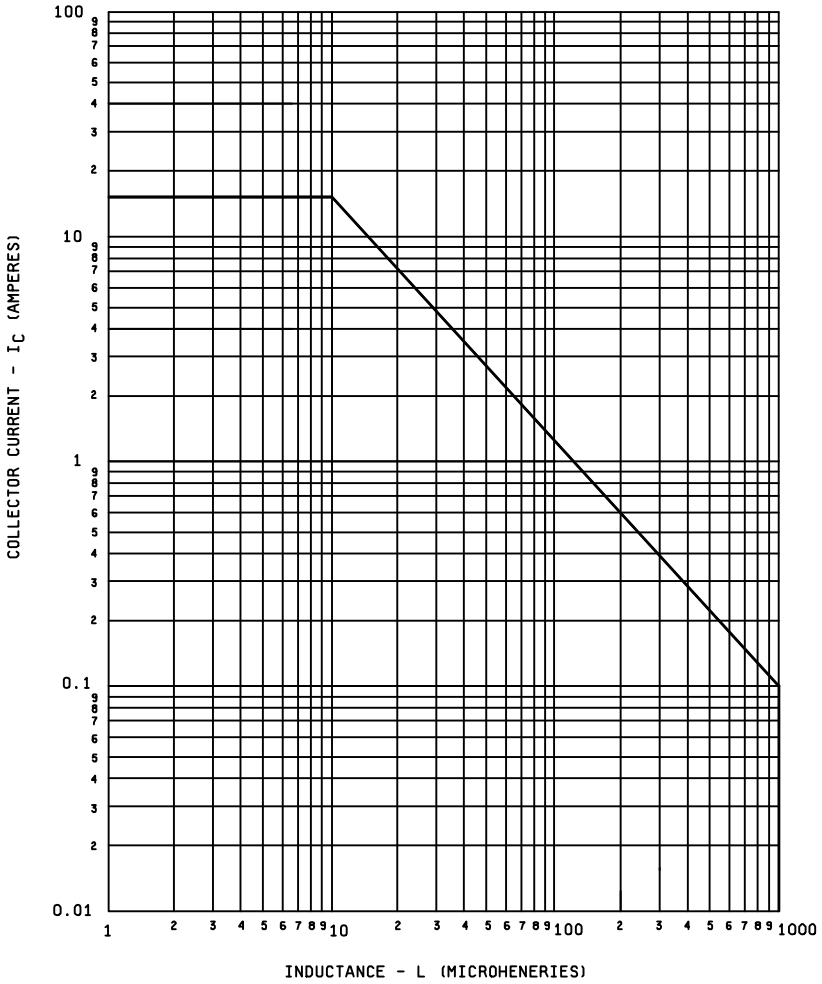
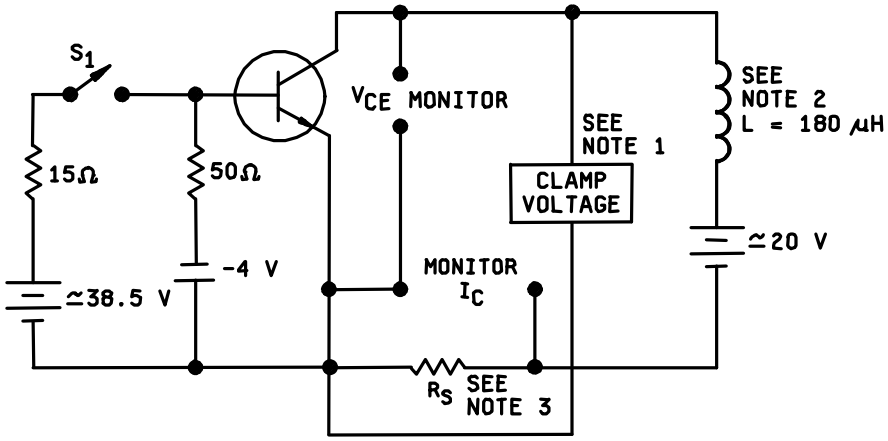


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 180 µH at 8 A with a maximum dc resistance of 0.05 ohm. For reference only: 2 each Miller type 7827 in parallel, or equivalent.
3. $R_S \leq .1$ ohm, 12 W, 1 percent tolerance maximum, (noninductive).

Procedure:
1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if clamp voltage not reached and maintained until current reaches zero.
3. Perform specified end-point tests.

FIGURE 5. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.2.1).

b. The lead finish as specified (see 3.4.1).

c. Type designation and quality assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List QML No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Air Force - 11
NASA – NA
DLA - CC

Review activities:
Air Force – 13, 19, 99

Preparing activity:
DLA - CC

(Project 5961-2075)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/525B	2. DOCUMENT DATE 990810

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER TYPES 2N6546, 2N6547 JAN, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD Form 1426, Feb 1999 (EG) Previous editions are obsolete WHS/DIOR, Feb 99

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